

Quiz 3

STAT 109: Introductory Biostatistics

Quiz 3

Quiz 3 Practice Problems

These are practice problems covering:

- **Lectures 3–4:** sample spaces, outcomes, events, and probability calculations when the outcomes are equally likely
 - **Lecture 5:** addition rule
 - **Lab 2 (R code by hand):** types, vectors, matrices, bracketing, `length()`, `dim()`, and basic `for` loops + simulation structure
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Instructions

- Work **without notes** first; then check your answers. I'll post my solutions to Lab 2 after the due date.
 - For **R questions**, write **valid R code** exactly as you would type it (by hand).
 - The in-class quiz in lab on Thursday will be created from these practice problems.
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Part A: Vocabulary and notation (Lecture 3)

A1. Identify terms for a trial

Consider the trial of rolling a single six-sided die one time.

1. Write the sample space S .
 2. Give one outcome.
 3. Define an event A and list the outcomes in A .
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A2. Set language in words

Suppose a trial has sample space S and two events A and B .

1. In words, what does “ A and B ” mean?
 2. In words, what does “ A or B ” mean?
 3. In words, what does “not A ” mean?
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A3. Translate probability notation to English

Write a one-sentence interpretation for each:

1. $P(A) = 0.7$
 2. $P(A \text{ and } B) = 0.2$
 3. $P(\text{not } A) = 0.1$
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Part B: Equally likely outcomes and careful counting (Lectures 3–4)

B1. Equally likely outcomes rule

State the **probability rule** for an event A if the outcomes in S are equally likely:

$$P(A) = \quad ?$$

Be sure to define what $|A|$ and $|S|$ mean in words.

B2. A probability rule

State the **condition** that must be true for the probability rule $P(A) = |A|/|S|$ to be valid.

B3. Two coin tosses

Suppose a fair coin is tossed **two times**.

1. List the sample space S so that all outcomes are equally likely.
 2. Let A be the event “at least one tail.” List the outcomes in A .
 3. Compute $P(A)$.
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B4. Two dice (sum equals 11)

Suppose a fair die is rolled **twice**.

1. How many outcomes are in the **sample space** S written so that all outcomes are equally likely?
 2. List the outcomes from S in the event $C =$ “the sum is 11.”
 3. Compute $P(C)$.
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B5. Random babies (Lecture 4)

Three babies are randomly returned to three mothers (assume all pairings are equally likely).

1. How many outcomes are in the sample space?
 2. Let A be the event “all 3 mothers get the correct baby.” Compute $P(A)$.
 3. Let B be the event “none of the mothers get the correct baby.” Compute $P(B)$.
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Part C: Addition rule and special cases (Lecture 5)

C1. State the addition rule

State the **Addition Rule** for any two events A and B .

C2. Backpacks and hats

A class has 65 students:

- 40 have backpacks
- 10 have hats
- 5 have **both** a backpack and a hat

Let A be the event a randomly selected student “has a backpack” and B be “has a hat.”

1. Compute $P(A)$ and $P(B)$.
 2. Compute $P(A \text{ and } B)$.
 3. Use the addition rule to compute $P(A \text{ or } B)$.
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C3. Complement rule (Special Case of Addition Rule)

If $P(\text{rains at least once in July}) = 0.91$, compute:

$$P(\text{no rain in July})$$

C4. Non-overlapping events (special case)

Suppose A and B are **non-overlapping** (mutually exclusive) events.

1. What does “non-overlapping” mean in words?
 2. What does it imply about $P(A \text{ and } B)$?
 3. Write the simplified formula for $P(A \text{ or } B)$ in this case.
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Part D: R code by hand (Lab 2)

D1. Types and `class()`

Write R code that:

1. creates `x_num` as the numeric value of 0,
 2. creates `x_int` as an integer (use the L),
 3. prints the class of each.
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D2. Vectors, length, and scalar operations

Write R code that:

1. creates a numeric vector `v` that contains the values 2, 4, 6, 8 and 10.
 2. prints the length of `v`
 3. creates a new vector called `v_plus_one` that is `v` with 1 added to each of its values.
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D3. Bracketing practice (vectors)

Using `v <- c(2, 4, 6, 8, 10)`, write R code for each:

1. the 3rd element
 2. the 2nd through 4th elements
 3. everything except the 3rd element
 4. all elements greater than 5
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D4. Matrices and `dim()`

Write R code that:

1. creates a 2x3 matrix `m` filled by column using `1:6`,
 2. prints the dimensions of `m`
 3. extracts the value of `m` in the second row and first column
 4. extracts the first row
 5. extracts the second column
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D5. A simple `for` loop with pre-allocation

Write R code that uses a `for` loop to fill a vector named `squares` with the squares of 1 through 5:

Expected result for `squares`

```
[1] 1 4 9 16 25
```

D6. Simulation structure (`for` loop + `mean()`)

Write R code to estimate the probability that a fair die roll is even using simulation:- simulate `n <- 10000` die rolls using `sample(1:6, size = n, replace = TRUE)` - compute the estimated probability with `mean(...)` Your code should create objects named `n`, `rolls`, and `p_even`.